



New Thermally and Power-Efficient PolarFire® SoC FPGAs

Twice the Performance, Half the Power

Edge compute systems need compact programmable, power-efficient devices and a small enough thermal footprint to eliminate fans and other heat mitigation while providing robust compute horsepower. Our PolarFire® SoC FPGAs have solved this challenge by delivering optimized power efficiency, with a hardened, real-time, Linux®-capable RISC-V based embedded microprocessor subsystem.

PolarFire SoC FPGAs offer the industry's only hardened application class RISC-V architecture-based processor complex with 2 megabytes (MB) of L2 cache and Low-Power DDR4 (LPDDR4) memory support, with industry-leading PolarFire FPGA fabric.

Overcome power, system size, cost and security challenges across all kinds of applications including smart embedded vision applications and thermally constrained automotive, industrial automation, communications, defense and IoT systems where neither power nor performance can be compromised.

Key Features

- Hardened, real-time RISC-V based microprocessors
- 55% lower power at 8,000 CoreMarks
- Product densities from 460K Logic Elements (LEs)
- 12.7G transceivers, multiprotocol 10Gb support
- Smallest form factor 11x11 mm packages

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